

CODE SPORT



In this month's column, we discuss a few computer science interview questions.

As we reach the end of 2016, we reflect briefly on some of the technical trends of this year. 2016 has been the year when AI, machine learning, IoT, and augmented virtual reality have become the buzz words within the tech community. It is also the year in which Big Data technology started stabilising and the hype started dying down as realistic implementations and adoption gathered momentum. On-the-fly, in-the-moment content experience has become the norm with Snapchat, Twitch, Periscope, Facebook Live Streaming, etc. Virtual augmented reality became a popular topic with the release of the Pokémon Go game. One question I frequently get asked, especially by our student readers, is, "What are the hot technologies I should read up and work on?" Well, there is no easy answer to that. I strongly believe that one should focus on gaining strong computer science fundamentals, first. Building a solid foundation on computer science basics such as data structures, algorithms and operating systems enables you to make your own choice on specialising in any area that is of interest to you, be it machine learning, AI, IoT or augmented reality. The reason I advocate that students get their fundamentals clear is because the specific areas that are considered 'hot' for a while can change over time, but each of these areas requires you to have sufficient knowledge in the basics of computer science, which hardly changes over time. Though this may sound trite, I still would advocate that our student readers make sure that they develop a solid foundation in algorithms, data structures and OSs. Then one can decide to develop expertise in a specific area, whether it is cloud computing, Big Data, IoT or machine learning.

Another question that I get asked frequently by student readers is which programming language they should develop expertise in. Again, there is no single right answer. My suggestion is that one should have

adequate proficiency in a systems programming language like C, a popular programming language like Python and any other languages that you need to do the job on hand! The programming language is nothing but a tool to achieve a particular task. Given the enormous self-help resources available on the Web today, especially Stack Exchange, it is not difficult to venture into a new programming language if your current task warrants it. In fact, the list of popular programming languages has changed little over the last one year, as you can see at <http://www.tiobe.com/tiobe-index/>. Java, C, C++ and Python still continue to rank among the top popular programming languages. One of the interesting things to note is the increasing popularity of the language 'Go', which is perhaps due to the interest in cloud programming — in particular, the Docker containers that use Go.

As is our regular practice, we close this year with a medley of computer science interview questions. We focus on a wide set of topics including algorithms, operating systems and machine learning.

1. What is the difference between using containers and virtual machines for resource control?
When would you prefer containers over virtual machines and vice versa?
2. You are given an array A of N integers and a number K. You need to find out how many times, if at all, K appears in array A. What would be the time complexity of your solution if (a) the given array A is unsorted, and (b) the given array A is sorted?
3. Given an array A of N characters, can you find out what the first character is in the array A that does not repeat itself?
4. What are virtual functions in C++? How are virtual functions implemented?
5. What are strongly typed languages and weakly typed languages? Can you give examples for each?